

Applied Regression Homework 1 225040015

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1 Introduction

This report presents the conceptual answers and applied analyses for Homework 1. All numerical results below are based on the provided computations.

2 Conceptual

1. (a)–(d)

- (a) **Flexible** is better: n large, p small, low overfitting risk, and flexible models can approximate the true relationship.
- (b) **Inflexible** is better: p large, n small, flexible models have high variance and overfit.
- (c) **Flexible** is better: the relationship is highly nonlinear and needs flexibility to capture curvature.
- (d) **Inflexible** is better: noise variance is high, so flexible models chase noise and increase variance.

3. Bias–Variance

(a) Shape of curves (low $\rightarrow$ high flexibility).

- Bias²: monotonically decreasing.
- Variance: monotonically increasing.
- Training error: monotonically decreasing.
- Test error: decreases first, then increases (U-shaped).
- Bayes error: flat (constant).

(b) **Explanation.** Increasing flexibility fits the training data more closely, lowering training error and bias. However, it also makes the model more sensitive to sample fluctuations, increasing variance. Test error combines bias², variance, and irreducible error, so it falls initially (bias reduction dominates) and then rises when variance dominates. Bayes error depends only on noise and is unrelated to model choice.

3 Applied 8: College Dataset

(a) Read data

```
college <- read.csv("College.csv")
```

(b) Set row names and drop school name column

```
rownames(college) <- college[,1]
college <- college[,-1]
```

(c)(i) Summary statistics

Table 1 reports the minimum, quartiles, median, mean, and maximum of each numeric variable.

Table 1: Summary of numeric variables in the College dataset

Variable	Min	Q1	Median	Mean	Q3	Max
Apps	81.000	776.000	1,558.000	3,001.640	3,624.000	48,094.000
Accept	72.000	604.000	1,110.000	2,018.800	2,424.000	26,330.000
Enroll	35.000	242.000	434.000	779.973	902.000	6,392.000
Top10perc	1.000	15.000	23.000	27.559	35.000	96.000
Top25perc	9.000	41.000	54.000	55.797	69.000	100.000
F.Undergrad	139.000	992.000	1,707.000	3,699.910	4,005.000	31,643.000
P.Undergrad	1.000	95.000	353.000	855.299	967.000	21,836.000
Outstate	2,340.000	7,320.000	9,990.000	10,440.700	12,925.000	21,700.000
Room.Board	1,780.000	3,597.000	4,200.000	4,357.530	5,050.000	8,124.000
Books	96.000	470.000	500.000	549.381	600.000	2,340.000
Personal	250.000	850.000	1,200.000	1,340.640	1,700.000	6,800.000
PhD	8.000	62.000	75.000	72.660	85.000	103.000
Terminal	24.000	71.000	82.000	79.703	92.000	100.000
S.F.Ratio	2.500	11.500	13.600	14.090	16.500	39.800
perc.alumni	0.000	13.000	21.000	22.744	31.000	64.000
Expend	3,186.000	6,751.000	8,377.000	9,660.170	10,830.000	56,233.000
Grad.Rate	10.000	53.000	65.000	65.463	78.000	118.000

Private counts: Yes = 565, No = 212.

(c)(ii) Pair plots (first 10 variables)

From the scatterplot matrix:

- **Apps**--**Accept**--**Enroll**--**F.Undergrad** show strong positive correlations.
- **Top10perc**--**Top25perc** shows a strong positive correlation.
- **Outstate** is positively related to **Expend** and **Top10perc**.

(c)(iii) Outstate vs Private

Private schools have clearly higher out-of-state tuition. Table 2 summarizes **Outstate** by group.

Table 2: Outstate tuition by group

Group	Min	Q1	Median	Q3	Max
Private=Yes	2,340.000	9,100.000	11,200.000	13,970.000	21,700.000
Private=No	2,580.000	5,366.000	6,609.000	7,844.000	15,732.000
Elite=Yes	5,224.000	12,219.000	16,950.000	18,411.500	20,100.000
Elite=No	2,340.000	7,070.000	9,556.000	12,155.000	21,700.000

(c)(iv) Elite indicator

Define **Elite** by `Top10perc > 50`. Counts: Yes = 78, No = 699. Elite schools have much higher **Outstate** tuition (median 16,950 vs 9,556).

(c)(v) Histograms

The distributions of **Apps**, **Enroll**, **F.Undergrad**, and **Expend** are strongly right-skewed. **Outstate** is moderately right-skewed, and **Grad.Rate** has a few values above 100.

(c)(vi) Summary

- Private and Elite schools tend to have higher out-of-state tuition.
- **Apps**--**Accept**--**Enroll**--**F.Undergrad** form the strongest size-related cluster.
- **Top10perc** and **Top25perc** are highly correlated, indicating consistent student quality measures.
- **Expend** is positively related to **Outstate** and **Top10perc**, suggesting resources and student quality move together.

Correlation highlights

Table 3 lists several variable pairs with high absolute correlations.

Table 3: Highly correlated variable pairs in the College dataset

Variable 1	Variable 2	Correlation
Enroll	F.Undergrad	0.965
Apps	Accept	0.944
Accept	Enroll	0.912
Top10perc	Top25perc	0.892
PhD	Terminal	0.850
Outstate	Expend	0.673
S.F.Ratio	Expend	-0.584

Enrollment scale variables are strongly positively correlated. Spending (**Expend**) is positively correlated with **Outstate**, while student-faculty ratio is negatively correlated with spending.

4 Applied 10: Boston Dataset

(a) Basic information

`dim(Boston)` = 506 rows and 14 columns (13 predictors + `medv`).

(b) Pairwise scatterplots

Key relationships from the scatterplot matrix:

- **rad**--**tax** is extremely positively correlated (about 0.91).
- **indus**--**nox** is positively correlated; **dis**--**nox** is negatively correlated.
- **rm**--**medv** is positively correlated; **lstat**--**medv** is strongly negatively correlated.
- **age**--**dis** is negatively correlated.

(c) Correlation with crime rate

Variables most correlated with `crim` (by absolute value):

Table 4: Correlation with `crim` in the Boston dataset

Variable	Correlation
rad	0.626
tax	0.583
lstat	0.456
nox	0.421
indus	0.407
medv	-0.388
b	-0.385
dis	-0.380

(d) Extremes and ranges

Maximum `crim` = 88.9762 (row 381), maximum `tax` = 711 (row 489), and maximum `ptratio` = 22 (row 355). Table 5 summarizes each variable's range.

Table 5: Variable ranges in the Boston dataset

Variable	Min	Max
crim	0.006	88.976
zn	0.000	100.000
indus	0.460	27.740
chas	0.000	1.000
nox	0.385	0.871
rm	3.561	8.780
age	2.900	100.000
dis	1.130	12.127
rad	1.000	24.000
tax	187.000	711.000
ptratio	12.600	22.000
b	0.320	396.900
lstat	1.730	37.970
medv	5.000	50.000

(e) Charles River

```
sum(Boston$chas == 1) = 35.
```

(f) Median pupil-teacher ratio

```
median(Boston$ptratio) = 19.05.
```

(g) Minimum-medv tract

The minimum `medv` is 5.0 at row 399. Its characteristics are listed in Table 6.

This tract has high `crim`, `nox`, `tax`, `ptratio`, and `lstat`, with low `rm` and `dis`, consistent with older, polluted, and less affluent areas.

Table 6: Variables for the minimum-`medv` tract

Variable	Value
<code>crim</code>	38.352
<code>zn</code>	0.000
<code>indus</code>	18.100
<code>chas</code>	0.000
<code>nox</code>	0.693
<code>rm</code>	5.453
<code>age</code>	100.000
<code>dis</code>	1.490
<code>rad</code>	24.000
<code>tax</code>	666.000
<code>ptratio</code>	20.200
<code>b</code>	396.900
<code>lstat</code>	30.590
<code>medv</code>	5.000

(h) Counts of large houses

`sum(Boston$rm > 7) = 64` and `sum(Boston$rm > 8) = 13`. Tracts with `rm > 8` tend to have higher housing prices and lower `lstat`.

5 Conclusion

- Conceptual results highlight the bias-variance tradeoff and when flexible vs. inflexible models are preferred.
- In the College data, private and Elite schools show higher out-of-state tuition, and enrollment-related variables are strongly correlated.
- In the Boston data, crime rate is positively related to accessibility and tax rate, and negatively related to housing prices and distance; the minimum-`medv` tract exhibits extreme values.